Interleaving recovers the full circuit distance of bivariate-bicycle codes

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Every non-interleaved single-ancilla syndrome-extraction circuit has circuit distance at most the distance of an extended code, which can fall below the code distance: at most 11 for the gross code $[[144, 12, 12]]$ and, as we show, at most 8 for $[[90, 8, 10]]$. We show that interleaving recovers the full code distance. We construct single-ancilla, interleaved circuits with circuit distance 12 for the gross code, answering an open question of Strikis, Browne and Beverland, and 10 for $[[90, 8, 10]]$, two above every non-interleaved circuit, and prove both under circuit-level depolarizing noise for every number of rounds. The method fixes an extended code and turns circuit design into a finite decision problem over its lightest logicals, solved exactly and certified with DRAT proofs; a direct search on the detector error model confirms both results. The circuits have periods of 18 and 16 time steps per round, against 8 for the IBM circuits.

Bivariate-bicycle (BB) codes [1] are leading candidates for low-overhead quantum memory; IBM's architecture is built on the gross code, with parameters $[[144, 12, 12]]$ [1, 2]. What a syndrome circuit delivers is its circuit distance d_{circ}, the minimum number of circuit faults that flip a logical observable without triggering any detector; at low physical error rates it governs the logical error rate.

No circuit for the gross code had been shown to reach $d_{\mathrm{circ}} = d = 12$; the IBM circuit, with 8 time steps per round, has $d_{\mathrm{circ}} \leq 10$ [1, 2]. Strikis, Browne and Beverland (SBB) [3] analysed single-ancilla circuits, in which each check is measured by one dedicated ancilla coupled by CNOTs. They append to the check matrix the "residual" errors that an ancilla fault spreads onto the data, obtaining an extended code of distance d_{ext}, and show that $d_{\mathrm{circ}} \geq d_{\mathrm{ext}}$ for every such circuit (their Prop. 3), while $d_{\mathrm{circ}} \leq d_{\mathrm{ext}}$ when the circuit is *non-interleaved* in their sense, i.e. when all X-type CNOTs precede all overlapping Z-type CNOTs or vice versa (their Sec. II A and Cor. 1). Since every CNOT order gives $d_{\mathrm{ext}} \leq 11$ for the gross code, non-interleaved circuits are capped at 11 (their Prop. 4); their depth-8 circuit (their Fig. 12, hereafter "Fig. 12") was conjectured to reach it. They left open "whether any interleaved circuit can achieve circuit distance 12" [3]. For the $[[90, 8, 10]]$ code of the same family the best known value is 8: the IBM circuit has $d_{\mathrm{circ}} \leq 8$ [1], proven exact by MaxSAT [4].

We answer this question in the affirmative and show that the gap between Prop. 3 and Cor. 1 can be wider than one. For the gross code we construct a single-ancilla, CNOT-based, interleaved circuit and prove $d_{\mathrm{circ}} = 12$ for every number of rounds (Theorem 1). For $[[90, 8, 10]]$ every choice of CNOT orders gives $d_{\mathrm{ext}} \leq 8$, so every non-interleaved circuit has $d_{\mathrm{circ}} \leq 8$; we construct an interleaved circuit with $d_{\mathrm{circ}} = 10 = d$ (Theorem 2). The idea (Fig. 1) is to fix an extended code and choose CNOT orders and times such that none of its lightest logicals can be realised by undetected circuit faults. This turns circuit design into a finite decision problem that we solve exactly and certify, a tool for single-ancilla circuits in general. Our computation of d_{ext} also proves SBB's conjecture that Fig. 12 has $d_{\mathrm{circ}} = 11$ (Corollary 1). The new circuits run with periods of 18 and 16 time steps per round instead of 8; we discuss this cost at the end.

Exact circuit distances are known for hypergraph-product codes [5], for the smallest BB circuit [6], and as certificates for other qLDPC constructions [7, 8]. Shaw and Terhal's morphing circuit with the same qubit count acts on a $[[144, 12, 12]]$ code that they call "a second 'gross' code", with $d_{\mathrm{circ}} \leq 12$ [9]; a morphing idling circuit for the gross code is "believed to have circuit distance 12" [10]. We note that this end-cycle code is not equivalent to the gross code, since its minimum stabiliser weight is 9 rather than 6. For $[[90, 8, 10]]$ the upper bound for the unchanged IBM circuit returns to 10 under infinitely Z-biased ancilla noise [11]; we keep depolarizing noise, change the circuit, and prove the value. Using SBB's lower bound via non-interleaved counterparts to screen interleaved schedules was proposed in Ref. [12]; for the gross code that bound gives at most 11. Check-dependent CNOT orders are an established tool [3, 13, 14].

Setting. We consider memory experiments in the Z or X basis: data initialisation, R rounds of syndrome extraction and transversal readout. Each check has one ancilla that is reset, coupled by six CNOTs and measured once per round; rounds of different checks may overlap in time (pipelining). Faults are the single-operation Pauli components of standard circuit-level depolarizing noise (SD6 [15]). Faults with identical detector and observable effects are merged into one detector-error-model (DEM) mechanism [16], and d_{circ} counts distinct mechanisms. For a check with CNOT order $(q_1, \ldots, q_6)$, an ancilla fault after the k-th CNOT deposits the residual $\{q_{k+1}, \ldots, q_6\}$ on the data; the extended code appends all residual columns (Hs, Ls) to the data columns (H: check matrix, L: logical operators). Every single fault, propagated to the readout, has a data image that is zero or one extended-code column (SBB, Table I; proved for our template in SM S5). For Fig. 12, $d_{\mathrm{ext}} = 11$ in both bases, with exactly 4224 weight-11 extended logicals per

basis, all realisable in Fig. 12 by eleven undetected faults.

Key idea. Interleaving decouples d_{circ} from d_{ext} in one direction only: $d_{\mathrm{circ}} \geq d_{\mathrm{ext}}$ still holds, but d_{circ} can exceed it. We say that an extended logical *lifts* if one can choose, for each of its columns, a circuit fault with that image such that the chosen faults trigger no detector; a lift is an undetected logical error of the same weight. Reaching $d_{\mathrm{circ}} = d$ requires an extended code of the largest possible distance and a timing under which none of its lightest logicals lifts. The first requirement is met by an invariance: up to the check itself, residuals of length 5, 4, 3 and 2 equal $\{q_1\}$, $\{q_1, q_2\}$, the complement of $\{q_1, q_2, q_3\}$ and $\{q_5, q_6\}$, so the extended code depends only on these three unordered sets per check. Reversal and the swaps s_{12}, s_{56} of positions 1,2 and 5,6 preserve them; all eight orders they generate share the extended code of Fig. 12, its distance 11 and its 4224 weight-11 logicals (Fig. 1a).

The second requirement concerns time. Reversing an order keeps the extended code but changes which fault realises each residual column and when its spread reaches the neighbouring checks (Fig. 1b). In Fig. 12 a hook column $\{q_5, q_6\}$ is produced after the fourth CNOT and lands after the neighbouring Z checks have read those qubits, so every data error can be placed after its reads. After reversal, the same column comes only from a fault between the second and third CNOT, whose four-qubit spread reaches one neighbour after its readout and another before it; along a cycle of checks, each seeing two errors of the logical, the early and late constraints leave one check with odd parity, so a detector fires. For all 360 logicals of this "ten data plus one hook" form, the lift fails at $R = 1$, and every irreducible conflicting set of detectors (one whose constraints alone forbid the lift, but no proper subset does) has between 4 and 10 detectors; the smallest such set has 4, 5 or 9 detectors. With the Fig. 12 orders themselves blocking is impossible: every retiming family of Table S3 is obstructed (End Matter).

Construction. We searched gross-code schedules symmetric under the 24 colour-preserving translations, choosing for each of the six (type, colour) classes one of the eight orders and its CNOT times with a SAT solver, under legality, determinism and the blocking of every weight-11 extended logical in a bulk Pauli-frame model (SM S12); candidates were then certified exactly. The result (Table I; timing diagram in SM S2) measures the gross code itself, is legal, deterministic, pipelined and interleaved, and every check's order differs from that of Fig. 12. A second period-18 circuit and a period-21 circuit in which all ancillas share one frame have the same distance. For $[[90, 8, 10]]$ the same kind of search over fully translation-invariant schedules gave the circuit in the lower part of Table I, in which every data qubit alternates X- and Z-type CNOTs six times per period.

Theorem 1 *Under SD6 noise, the gross-code circuit of Table I has $d_{\mathrm{circ}} = 12$ in both memory bases for every number of rounds $R \geq 1$.*

The proof rests on two lemmas valid for every circuit of our template.

Lemma 1 (Reduction) *An undetected fault set of size w that flips a logical observable yields an extended-code logical of weight at most w, formed by its nonzero fault images.*

Proof. Add one observable per row of H and L on the final data readout. The observable of row H_i equals the sum of all detectors of check i, which telescope from the first measurement to the final readout. For an undetected fault set the H parts of the images therefore sum to zero and the L parts to the logical effect; each image is zero or one column, and equal columns cancel. □

Lemma 2 (Light logicals) *An undetected logical of weight d_{ext} consists of d_{ext} faults whose images are distinct columns forming an extended logical. One of weight $d_{\mathrm{ext}} + 1$ consists either of $d_{\mathrm{ext}} + 1$ such faults, or of d_{ext} such faults forming an extended logical together with one fault of zero image.*

The proof is a two-line count (End Matter). Deciding $d_{\mathrm{circ}} \geq d_{\mathrm{ext}} + 2$ is thus a finite problem: decide which extended logicals of weight d_{ext}, with or without one zero-image fault, and of weight $d_{\mathrm{ext}} + 1$ lift.

For the gross circuit, Lemma 1 with $d_{\mathrm{ext}} = 11$ excludes weight ≤ 10, and only weight 11 remains.

Lemma 3 (Finite rounds) *For $R = 1, \ldots, 32$ and both bases, none of the 4224 weight-11 extended logicals of the gross circuit lifts.*

We verified this by an exact computation, a CP-SAT model per logical that chooses one DEM mechanism per column subject to every detector sum vanishing, with two independent implementations; the same program finds a lift for all 4224 logicals in Fig. 12 (positive control). Each infeasible instance was also encoded as CNF; batches combined by a selector disjunction were refuted with DRAT proofs verified with drat-trim, whose LRAT cores were verified with cake_lpr (SM S6). The inputs, $d_{\mathrm{ext}} = 11$ and the list of 4224 logicals, are confirmed by two independent exhaustive searches rather than by proof certificates.

Lemma 4 (All rounds) *For the gross circuit, a weight-11 undetected logical for some R implies one for some $R' \leq 32$.*

A fault-free window of 52 layers would split such a logical into two undetected parts, one of weight ≤ 10; deleting fault-free periods then bounds the number of rounds (End Matter). A weight-12 data logical, applied as flips after initialisation, triggers no detector and flips the observable, which completes the proof of Theorem 1.

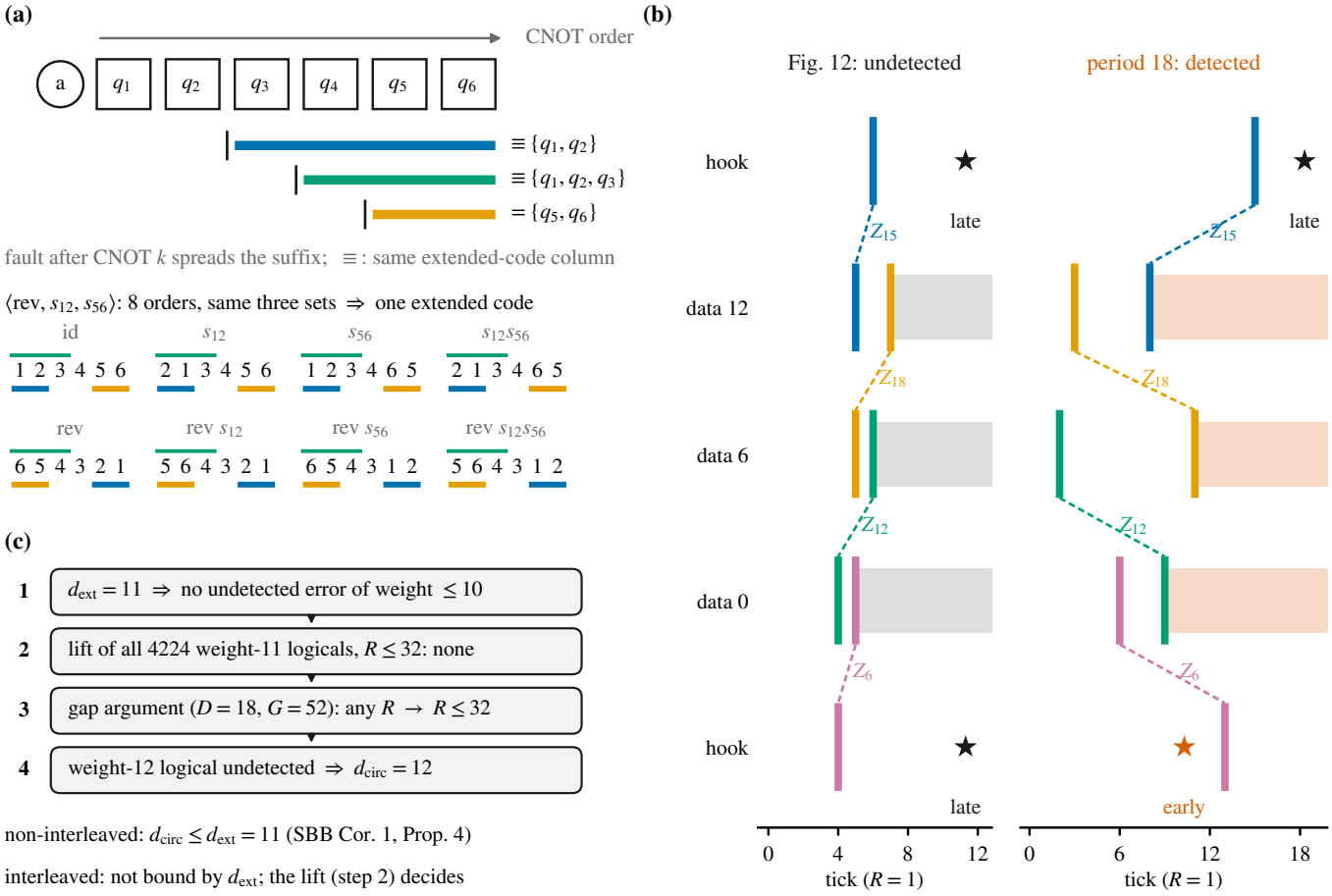


FIG. 1. The idea, on the gross code. (a) Residuals of length 4, 3 and 2 give the same extended-code columns as $\{q_1, q_2\}$, $\{q_1, q_2, q_3\}$ and $\{q_5, q_6\}$, so the eight orders generated by reversal and the swaps s_{12}, s_{56} share one extended code. (b) A weight-11 extended logical (ten data errors and the hook column $\{9, 25\}$; Z memory, $R = 1$). Bars: reads of the Z checks Z_{15}, Z_{18}, Z_{12}, Z_6; stars: hook arrival. Left, Fig. 12: the hook lands after both reads and no detector fires. Right, period-18 circuit: the same column comes from the spread $\{8, 91, 82, 85\}$ (equal to $\{9, 25\}$ up to the check), seen late by Z_{15} and early by Z_6; the forced timings (red) leave odd parity at Z_6. (c) The proof of Theorem 1.

Corollary 1 *The Fig. 12 circuit of SBB has* $d_{circ} = 11$ *in both bases for every* R*, as they conjectured.*

This is SBB's Prop. 3 and Cor. 1 combined with our value $d_{ext} = 11$.

Gap of two for $[[90, 8, 10]]$. For every assignment of per-check CNOT orders, the extended code of the X-check side contains a logical of weight at most 8, as we verified with a SAT encoding whose unsatisfiability is certified by a DRAT proof checked with drat-trim (SM S9). Hence $d_{ext} \leq 8 = d - 2$ for every single-ancilla circuit of this code, and every non-interleaved circuit has $d_{circ} \leq 8$ by SBB's Cor. 1, a value that is attained.

Theorem 2 *Under SD6 noise, the* $[[90, 8, 10]]$ *circuit of Table I has* $d_{circ} = 10$ *in both memory bases for every number of rounds* $R \geq 1$.

Proof sketch. Its orders give $d_{ext} = 8$, with exactly 585 extended logicals of weight 8 and 16830 of weight 9 per basis. By Lemmas 1 and 2, $d_{circ} \geq 10$ holds if none of these lifts, the weight-8 ones also together with one zero-image fault. For $R = 1, \ldots, 12$ and both bases none does, as found by three independent implementations. Rounds are separated in time, which reduces every R to $R \leq 12$, and a weight-10 data logical gives the upper bound (End Matter).

To our knowledge, the two circuits of this work are the first explicit single-ancilla circuits whose circuit distance exceeds the extended-code distance of every choice of CNOT orders. Both lower bounds were also checked without the extended code, by an exact branch-and-bound search on the full DEM (an independent implementation of the same family of methods as SM S4): for the gross circuit there is no undetected logical of weight ≤ 11 at $R = 1, 2, 3$; for the $[[90, 8, 10]]$ circuit none of weight ≤ 9 for $R \leq 12$, which with round splitting proves Theorem 2 without extended-code input (SM S10).

TABLE I. The two circuits. Gross code (period 18): checks are grouped by type and colour $(x+y)$ mod 3; all 24 checks of a class share the entries; steps are modulo 18, and a window passing step 17 continues in the next period; last column: map from the Fig. 12 order. $[[90,8,10]]$ (period 16): every check of a type shares the entries; steps 11, 13, 14 contain no CNOT. Labels: $A = x^3 + y + y^2$, $B = y^3 + x + x^2$ (gross) and $A = x^9 + y + y^2$, $B = 1 + x^2 + x^7$ ($[[90,8,10]]$) [1], $A \mapsto A_1A_2A_3$, $B \mapsto B_1B_2B_3$.

class	reset	CNOT order	CNOT steps	meas.	map
X_0	17	$A_1A_2A_3B_3B_1B_2$	0, 1, 2, 8, 9, 15	16	rev s_{56}
X_1	15	$A_1A_2A_3B_1B_3B_2$	16, 3, 4, 5, 10, 13	14	rev
X_2	17	$A_3A_1A_2B_3B_1B_2$	0, 1, 5, 6, 7, 15	16	rev s_{56}
Z_0	17	$B_2B_1B_3A_2A_1A_3$	0, 7, 8, 9, 13, 14	15	rev s_{56}
Z_1	2	$B_2B_3B_1A_3A_2A_1$	3, 4, 9, 10, 11, 17	0	rev $s_{12}s_{56}$
Z_2	16	$B_2B_1B_3A_3A_2A_1$	17, 5, 6, 12, 13, 14	15	rev s_{56}
X	0	$A_2A_1A_3B_2B_3B_1$	1, 3, 5, 6, 8, 10	15	–
Z	0	$B_1B_3B_2A_3A_1A_2$	2, 4, 6, 7, 9, 12	15	–

Optimality of the gross period. In a bulk Pauli-frame model of lifting, no gross-code schedule symmetric under the 24 colour-preserving translations with orders in the class of Fig. 12 has period below 18, or below 21 if all ancillas share one frame; a model lift is a genuine error for logicals with one length-2 residual and was confirmed for all 991 sampled others (End Matter). Among all 24 colourings by cosets of translation subgroups of index 2, 3, 4, 6, 8 or 9, with one order type per colour, only the class of Fig. 12 has $d_{\text{ext}} \geq 11$.

Discussion. SBB's obstruction is specific to non-interleaved circuits: both codes admit single-ancilla circuits that preserve their full distance, and for $[[90,8,10]]$ interleaving gains two units over every non-interleaved circuit. Fixing the extended code and searching over orders and timing turns an open-ended circuit search into a lift problem against finite lists of extended logicals of weight d_{ext} and $d_{\text{ext}}+1$, which is decided exactly; it applies to every single-ancilla circuit whose distance is limited by SBB's Cor. 1.

Cost and outlook. The gross circuit uses the qubits of the IBM and SBB circuits but 18 time steps per round instead of 8, with 3168 idle locations per round instead of 288 and 2.8 times the total fault probability. Under the same SD6 noise and identical BP-OSD and Relay-BP decoders, it is worse than the IBM circuit at every sampled $p \geq 10^{-3}$ (about 1.5×10^3 times at $p = 2\times10^{-3}$, at least 3 times at 10^{-3}), while at matched fault mass the circuits perform alike. Rescaling only the idle channel, $p_{\text{idle}} = \lambda p$, shows where the structure pays off: at $\lambda = 0$ the period-18 circuit beats Fig. 12 at $p = 4\times10^{-3}$ (ratio 0.37 [0.28, 0.49] with BP-OSD) and is statistically indistinguishable from the IBM circuit at $p = 3$–4×10^{-3}, whereas at $\lambda = 0.1$ it is worse than Fig. 12 by 2.3–3 times per round (SM S13). The $[[90,8,10]]$ circuit has period 16, three steps of which contain no CNOT, against 8 for the IBM circuit; its logical error rate has not been evaluated. Shorter periods are therefore the natural next goal: can $d_{\text{circ}} = 12$ and $d_{\text{circ}} = 10$ be reached at period 8 by less symmetric schedules? Correlated faults, leakage and CZ-native compilations are outside the scope of this work.

Data availability. The circuit specifications, the lists of extended logicals used in the lift computations (4224 of weight 11 for the gross circuit; 585 of weight 8 and 16830 of weight 9 for the $[[90,8,10]]$ circuit, each basis), the witnesses, the certificates (clause sets of the unsatisfiable instances) and the reproduction scripts are available at [Zenodo DOI].

AI use. This research was carried out with AI agents (Claude, Anthropic). The agents proposed hypotheses, wrote and ran the search and verification code, worked out the constructions and proofs, and independently rechecked each claim reported here, using separately written code for every computation. The author directed the project, made all decisions at its decision points, and takes full responsibility for the content. In line with APS policy, no AI system is listed as an author.

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[1] S. Bravyi, A. W. Cross, J. M. Gambetta, D. Maslov, P. Rall, and T. J. Yoder, High-threshold and low-overhead fault-tolerant quantum memory, Nature **627**, 778 (2024), arXiv:2308.07915.
[2] T. J. Yoder, E. Schoute, P. Rall, E. Pritchett, J. M. Gambetta, A. W. Cross, M. Carroll, and M. E. Beverland, Tour de gross: A modular quantum computer based on bivariate bicycle codes, arXiv:2506.03094 (2025).
[3] A. Strikis, D. E. Browne, and M. E. Beverland, High-performance syndrome extraction circuits for quantum codes, arXiv:2603.05481 (v2, 2026).
[4] L. Aghababaie Beni, O. Higgott, and N. Shutty, Tesseract: A search-based decoder for quantum error correction, arXiv:2503.10988 (v2, 2025).
[5] A. G. Manes and J. Claes, Distance-preserving stabilizer measurements in hypergraph product codes, Quantum **9**, 1618 (2025), arXiv:2308.15520.

[6] M. Webster, A. Jacob, and O. Higgott, Distance-finding algorithms for quantum codes and circuits, arXiv:2603.22532 (v2, 2026).
[7] Z. Liang, B. Gu, Y.-A. Chen, J. Eisert, and Z. Wang, Ultra-high-distance quantum memories from amplified qLDPC codes, arXiv:2609.37231 (2026).
[8] Z. Sun and Z. Cai, Layer codes as quantum memories: syndrome extraction, thresholds and idle robustness, arXiv:2609.39522 (v2, 2026).
[9] M. H. Shaw and B. M. Terhal, Lowering connectivity requirements for bivariate bicycle codes using morphing circuits, Phys. Rev. Lett. **134**, 090602 (2025), arXiv:2407.16336.
[10] A. Wills, M. E. Beverland, L. S. Bishop, J. M. Gambetta, P. Rall, V. Siddhu, and A. W. Cross, Concatenating algebraic codes over high-rate quantum LDPC codes, arXiv:2605.21898 (v2, 2026).
[11] R. Bi, K. Chang, and S. Puri, Untangling QLDPC codes with biased noise ancilla, arXiv:2606.30592 (2026).
[12] M. T. P. Nguyen, M. Rimbach-Russ, and S. Bosco, Disassembling qLDPC codes for depth-optimal parity-check circuits, arXiv:2608.19917 (v2, 2026).
[13] Y. Tomita and K. M. Svore, Low-distance surface codes under realistic quantum noise, Phys. Rev. A **90**, 062320 (2014), arXiv:1404.3747.
[14] G. Kishony and A. Fowler, Color code off-the-hook: avoiding hook errors with a single auxiliary per plaquette, arXiv:2603.28852 (2026).
[15] C. Gidney, M. Newman, A. Fowler, and M. Broughton, A fault-tolerant honeycomb memory, Quantum **5**, 605 (2021), arXiv:2108.10457.
[16] C. Gidney, Stim: a fast stabilizer circuit simulator, Quantum **5**, 497 (2021), arXiv:2103.02202.
[17] M. H. Shaw and B. M. Terhal, Optimising quantum error correction using morphing circuits, arXiv:2604.09797 (v2, 2026).

END MATTER

Proof of Lemma 2. Let w faults form an undetected logical, z of them with zero image and c pairs with equal nonzero images, which cancel. By Lemma 1 the remaining distinct columns form an extended logical of weight $w - z - 2c \geq d_{\text{ext}}$. For $w = d_{\text{ext}}$ this forces $z = c = 0$; for $w = d_{\text{ext}} + 1$ it forces $c = 0$ and $z \leq 1$. □

Proof of Lemma 4. Let $D = 18$ be the period. After the last fault of a fault set at layer a, the data frame is constant from layer $a + D - 1$ on, because only ancillas still in flight can change it. If an undetected weight-11 logical had no fault in a window of $G = 2D + \Delta - 1 = 52$ layers, where $\Delta = 17$ is the longest reset-to-measurement span of a check, no detector could respond to both the faults before and after the window; both parts would be undetected, their logical effects would add to a nonzero value, and one part would have weight ≤ 10, contradicting Lemma 1. So a weight-11 logical spans at most 10×51 layers. The noisy circuit is periodic: inserting an 18-layer block at any layer x with $4 \leq x \leq 18(R-1)$ (layers numbered from 0) turns the $(R-1)$-round circuit into the R-round one. Deleting fault-free blocks before and after the logical leaves every detector it touches unchanged, which maps it into a circuit with at most 32 rounds (term-by-term count in SM S7). With Lemma 3 no weight-11 undetected logical exists for any R. □

Details for Theorem 2. The lists of 585 weight-8 and 16830 weight-9 extended logicals were obtained by two independent enumerations (a depth-first and a meet-in-the-middle search) with identical results. In this circuit the zero-image mechanisms that trigger detectors are exactly the ancilla readout flips, 45 per round; all zero-image mechanisms are admitted as candidates. The three lift implementations (a frontier dynamic program, a CP-SAT model and a per-logical SAT model) find no lift for $R = 1, \ldots, 12$ in both bases; as positive controls they find 315 weight-9 lifts for a nearby interleaved circuit with $d_{\text{circ}} = 9$, and 585 weight-8 lifts for a non-interleaved circuit with the same extended code. All ancillas are reset at step 0 and measured at step 15, so rounds are separated in time and the circuit is periodic. A minimal undetected logical with a fault-free round inside its span would split into two undetected parts, one lighter; a weight-9 logical therefore spans at most 9 rounds, and deleting fault-free rounds after the second and before the last maps it into a circuit with at most $2+9+1 = 12$ rounds. A weight-10 data logical, applied as flips after initialisation, fires no detector and flips the observable (SM S9).

Retimings of Fig. 12. Among periodic schedules of period ≤ 8 keeping the Fig. 12 orders there are four conflict-free ones up to translation, all with injection-verified weight-11 undetected errors for $R = 1, \ldots, 6$. For longer periods, in a bulk Pauli-frame model a data error is seen by a check at the first readout after it, and a logical is model-liftable if its errors can be timed so that every check sees an even number of them between consecutive readouts. A model lift of a logical with ten data errors and one length-2 residual is a genuine undetected error once the number of rounds is large enough (a one-dimensional Helly argument). Every legal circuit keeping the orders has such an error in each family of Table S3, as shown by SAT encodings restricted to these logicals, each accompanied by a DRAT proof (SM S11).

Optimality of the period. In the same model, among gross-code schedules symmetric under the 24 colour-preserving translations with orders in the class of Fig. 12, every legal schedule with period $P \leq 17$ has a model-liftable weight-11 logical, and so does every schedule with $P \leq 20$ in which all ancillas share one reset and measurement frame; the minimal periods are therefore 18 and 21 in this model. These encodings use all modelled logicals; the model-to-circuit step is proven for logicals with one length-2 residual and was confirmed on the exact detector error model in all 991 sampled cases with several residual columns. In the same model, invariance under a translation subgroup of order 12 already excludes $P \leq 11$, and

any non-trivial subgroup excludes $P = 8$; schedules without symmetry are open. The order class is exact: among the 24 colourings by cosets of translation subgroups of index 2, 3, 4, 6, 8 or 9, with one order type per colour, only the class of Fig. 12 has $d_{\mathrm{ext}} \geq 11$ in both bases (SM S12).

Supplemental Material

The sections follow the order of the main text. S1–S2 fix notation and the gross circuits; S3–S8 contain the proof of Theorem 1 and the general lemmas; S9 treats the $[[90, 8, 10]]$ code and Theorem 2; S10 the direct search on the detector error model; S11 Fig. 12 of SBB; S12 the search and the optimality statements; S13 the sensitivity of the logical error rate to idle noise; S14 the verification and the data.

S1. NOTATION AND NOISE MODEL

Code. The gross code [1] has $\ell = 12$, $m = 6$, $A = x^3 + y + y^2$, $B = y^3 + x + x^2$, $H_X = [A \,|\, B]$, $H_Z = [B^T \,|\, A^T]$; qubits and checks are indexed by $i = 6a + b$ for the monomial $x^a y^b$, with left and right blocks of 72 qubits each; $n = 144$, $k = 12$, $d = 12$. Term labels: $A_1 = x^3$, $A_2 = y$, $A_3 = y^2$, $B_1 = y^3$, $B_2 = x$, $B_3 = x^2$. The X check at (x, y) acts on left qubit $(x, y) + A_k$ and right qubit $(x, y) + B_k$; the Z check at (x, y) on left qubit $(x, y) - B_k$ and right qubit $(x, y) - A_k$. The colour of a check is $(x + y)$ mod 3; the 24 translations (a, b) with $a + b \equiv 0 \pmod 3$ preserve colours.

Circuit template. Each check has one ancilla. In every round it is reset, applies its six CNOTs in its order (ancilla as control for X checks, as target for Z checks) and is measured. A schedule assigns each check a reset step, six CNOT steps and a measurement step modulo the period P; windows may wrap around the period boundary, so that rounds of different checks overlap. A Z-basis memory initialises the data in $|0\rangle$, runs R rounds and reads all data in the Z basis; detectors are the first measurement of each Z check, the comparison of consecutive measurements of each Z check, and the comparison of the last measurement with the parity of the final readout. The X basis is analogous.

Noise and d_{circ}. SD6 [15]: after every CNOT a two-qubit depolarizing channel; after every reset a flip; before every measurement a flip; on every qubit idle in a time step a single-qubit depolarizing channel (also during reset and measurement layers). A fault is one Pauli component of one channel. Faults with identical detector and observable signatures are one DEM mechanism, and d_{circ} counts mechanisms. For lower bounds we use the CSS relaxation: in a Z-basis memory only Z-check detectors and the X part of each fault are kept; this can only lower the minimum. Upper bounds are checked on the full circuit. Lower bounds hold for any sub-model of SD6.

S2. THE CIRCUITS

Table I specifies circuit A completely: each of the 144 checks takes the reset step, order, CNOT steps and measurement step of its class. Its resets occur at steps $\{2, 15, 16, 17\}$ and its measurements at $\{0, 14, 15, 16\}$. We verified for circuits A, B and C that every check applies exactly six CNOTs to its support at distinct steps inside its window, that no data qubit takes part in two CNOTs at the same step, that the noiseless circuit fires no detector, and that all detectors are deterministic, for several R in both bases. Interleaving: in circuit A, over one period, 96 data qubits switch four times and 48 twice (cyclically) between X- and Z-type CNOTs.

Circuit B (Table S1) also has period 18; 24 of its checks keep the Fig. 12 order. Circuit C (Table S2) has period 21 and a common frame: all ancillas are reset at step 0 and measured at step 20, so rounds are separated in time. Both have $d_{\mathrm{circ}} = 12$ for every R in both bases (S6, S7).

For reference, the Fig. 12 orders in the same labelling are X_0: $B_1B_2B_3A_3A_2A_1$, X_1: $B_2B_3B_1A_3A_2A_1$, X_2: $B_1B_2B_3A_2A_1A_3$, Z_0: $A_1A_3A_2B_3B_1B_2$, Z_1: $A_2A_1A_3B_1B_2B_3$, Z_2: $A_2A_1A_3B_3B_1B_2$ (period 8, pipelined, non-interleaved).

TABLE S1. Circuit B (period 18). Columns as in Table I.

class	reset	CNOT order	CNOT steps	meas.	map
X_0	0	$B_2B_1B_3A_3A_2A_1$	1, 7, 8, 14, 15, 16	17	s_{12}
X_1	2	$B_2B_3B_1A_3A_2A_1$	3, 6, 11, 12, 13, 0	1	id
X_2	0	$B_2B_1B_3A_2A_1A_3$	1, 9, 10, 11, 15, 16	17	s_{12}
Z_0	1	$A_3A_1A_2B_3B_1B_2$	2, 3, 7, 8, 9, 17	0	s_{12}
Z_1	17	$A_1A_2A_3B_1B_3B_2$	0, 5, 6, 7, 12, 15	16	$s_{12}s_{56}$
Z_2	1	$A_1A_2A_3B_3B_1B_2$	2, 3, 4, 10, 11, 17	0	s_{12}

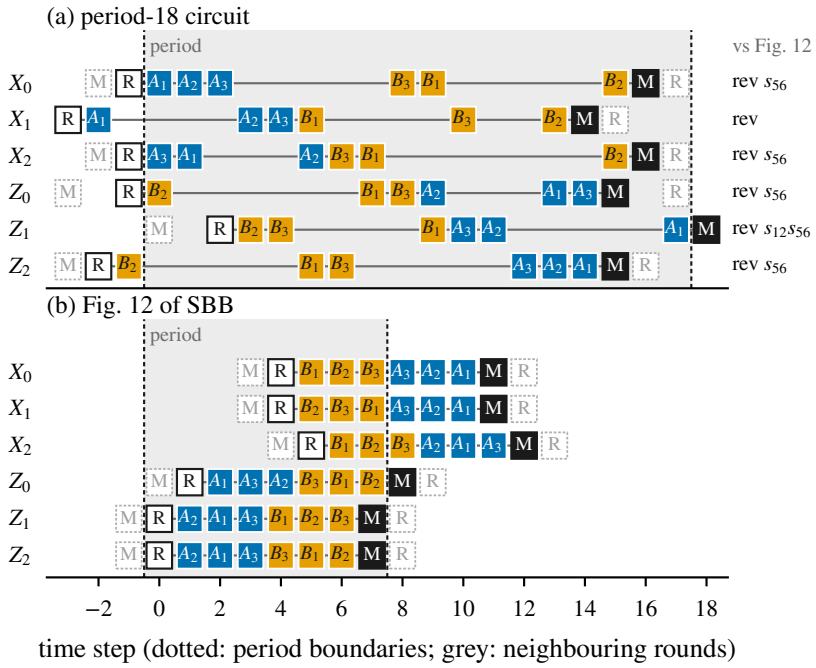


FIG. S1. One period of (a) the gross-code circuit of Table I and (b) Fig. 12 of SBB [3], on the same time scale. Rows: the six classes; boxes: CNOTs labelled by the term they implement; R, M: ancilla reset and measurement. Windows crossing the period boundary (dotted) continue into the next period. (b) has period 8 and is not interleaved: on every data qubit the X- and Z-type CNOTs of a round form two contiguous blocks.

TABLE S2. Circuit C (period 21, common frame: reset at 0, measurement at 20).

class	CNOT order	CNOT steps	map
X_0	$B_2B_1B_3A_3A_2A_1$	2, 7, 9, 15, 16, 17	s_{12}
X_1	$B_2B_3B_1A_3A_2A_1$	1, 8, 12, 13, 14, 17	id
X_2	$B_2B_1B_3A_2A_1A_3$	1, 10, 11, 12, 16, 18	s_{12}
Z_0	$A_3A_1A_2B_3B_1B_2$	3, 4, 8, 9, 10, 17	s_{12}
Z_1	$A_1A_2A_3B_1B_3B_2$	1, 6, 7, 8, 13, 14	$s_{12}s_{56}$
Z_2	$A_1A_2A_3B_3B_1B_2$	2, 4, 5, 11, 12, 19	s_{12}

S3. INVARIANCE OF THE EXTENDED CODE

Consider a Z-basis memory; only X errors matter. An X error on the ancilla of an X check after its k-th CNOT propagates to the suffix $\{q_{k+1}, \ldots, q_6\}$; Z-check CNOTs (data controls, ancilla target) do not change the X frame on the data. Each fault's final data X frame is therefore zero, a single data qubit, or a proper suffix of an X-check order, and its column is $(H_Z s, L_Z s)$. Multiplying by the check itself, which has zero column, a suffix of length 5 is equivalent to $\{q_1\}$ (a data column), of length 4 to $\{q_1, q_2\}$, of length 3 to the complement of $\{q_1, q_2, q_3\}$, and a suffix of length 2 is $\{q_5, q_6\}$. Hence the column set is determined by the three sets $\{q_1, q_2\}$, $\{q_1, q_2, q_3\}$, $\{q_5, q_6\}$ of every check. Reversal maps these to $\{q_5, q_6\}$, the last three and $\{q_1, q_2\}$, which produce the same columns, and s_{12}, s_{56} leave them unchanged. The X basis is symmetric. For circuits A, B and C we also computed the extended codes directly: both bases have 360 merged columns, and the column sets coincide with those of Fig. 12.

S4. THE EXTENDED CODE: $d_{\text{ext}} = 11$ AND THE 4224 WEIGHT-11 LOGICALS

The extended code of Fig. 12 has 360 merged columns per basis (144 of check-weight 3, 144 of weight 6, 72 of weight 9). The 24 colour-preserving translations map each check's order to the order of its image check and preserve

the logical classes (checked by solving $L \circ T = ML + NH$ with $\operatorname{rank} M = k$); the columns form 15 orbits.

Search. To decide whether $Hx = 0$, $Lx \neq 0$ has a solution with $|x| \leq W$ we use a complete depth-first search: a minimum-weight solution may be translated so that it contains the representative of its lowest orbit; the search branches on the columns covering an unsatisfied check row, excluding earlier siblings, so that the branches partition the search space; it prunes when the depth plus $\lceil |s|/w_{\max} \rceil$ exceeds W (s the current syndrome, $w_{\max}$ the largest column weight) and closes the last step by table look-up. The program was validated against unpruned meet-in-the-middle enumeration on smaller instances and by checking that split runs add up to unsplit runs.

Results. At $W = 10$ there is no solution in either basis ($\approx 2 \times 10^{10}$ nodes per basis), on two independently constructed instances, and under a second symmetry group of order 8 (45 orbits, $\approx 5 \times 10^{10}$ nodes per basis); weight 11 is attained. Hence $d_{\text{ext}} = 11$. At $W = 11$, two different search programs enumerate the same 4224 extended logicals per basis, all in free orbits of the 24-element group (176 orbits); 360 of them consist of ten data columns and one residual of length 2. Both $d_{\text{ext}} = 11$ and the completeness of the list of 4224 weight-11 logicals rest on two independent exhaustive searches; we did not obtain a checkable proof certificate for them (a direct CNF encoding is beyond CDCL solvers already at weight 6).

S5. PROOF OF LEMMA 1

We work in a Z-basis memory; the X basis follows by exchanging the roles of X and Z and of the two check types. In a Z-basis memory only X components matter. For a fault we track its *data X frame*: the set s of data qubits carrying an X error after the fault has been propagated through the rest of the circuit to the final data readout. Its *image* is the column $(H_Z s,\, L_Z s)$.

Gates of the template. An X-check ancilla is prepared in $|+\rangle$, is the control of its six CNOTs (data qubits as targets) and is measured in the X basis. A Z-check ancilla is prepared in $|0\rangle$, is the target of its six CNOTs (data qubits as controls) and is measured in the Z basis. Under a CNOT, an X on the control is copied to the target, an X on the target stays there, and Z errors never create X errors. Hence: (a) an X on a data qubit never leaves it (as a target it is not copied; as a control it is copied only onto Z-check ancillas, which are never controls and are measured and reset without acting back on the data); (b) an X on a Z-check ancilla never reaches the data; (c) an X on an X-check ancilla after its k-th CNOT of an instance is copied onto exactly the targets $q_{k+1}, \ldots, q_6$ of the remaining CNOTs of that instance and is then discarded by the next preparation (the X-basis measurement does not see it). These statements use only the direction of each gate, not the relative timing of different checks, so they hold for interleaved and pipelined schedules alike.

Single faults. We go through the SD6 locations. In each case s is empty, a single data qubit, or a suffix $\{q_k, \ldots, q_6\}$ of one X-check order.

1. Data qubits: an X or Y from idle noise, a flip after data preparation, or a flip before the final readout gives $s = \{q\}$ by (a); a Z gives $s = \emptyset$.

2. Z-check ancillas: preparation and measurement flips, idle errors and any error on the ancilla alone give $s = \emptyset$ by (b).

3. X-check ancillas: the preparation flip and the measurement flip are Z errors in this basis and give $s = \emptyset$; an X or Y from idle noise before the first CNOT gives the full support of the check, after the k-th CNOT the suffix $\{q_{k+1}, \ldots, q_6\}$, after the sixth CNOT $s = \emptyset$, by (c).

4. Two-qubit Pauli $P_a \otimes P_d$ after the k-th CNOT of an X check (ancilla a, data q_k): the X part of P_a gives $\{q_{k+1}, \ldots, q_6\}$ and the X part of P_d gives $\{q_k\}$, so $X_a I$ gives $\{q_{k+1}, \ldots, q_6\}$, $I X_d$ gives $\{q_k\}$, and $X_a X_d$ (or any combination with X parts on both) gives their union $\{q_k, q_{k+1}, \ldots, q_6\}$, the suffix starting at q_k.

5. Two-qubit Pauli after a CNOT of a Z check (data q, ancilla): the data part gives $\{q\}$ or $\emptyset$ by (a), the ancilla part gives $\emptyset$ by (b).

Images. A single data qubit gives a data column of the extended code. A suffix $\{q_k, \ldots, q_6\}$ with $2 \leq k \leq 6$ is a residual, i.e. one of the appended columns (for $k = 6$ it is the data column of q_6). The full support ($k = 1$) is the check itself, whose image is zero because H_Z and L_Z commute with the X stabilisers. So every single fault has image zero or exactly one extended-code column. Merging faults into DEM mechanisms does not change this, since merged faults have identical effects on all detectors and observables, including the augmented ones introduced below.

Telescoping. Augment the circuit with one observable for each row of H_Z and of L_Z, measured on the final data readout. For every Z check i, the sum of all its detectors (first measurement, all consecutive comparisons, final comparison) telescopes to the parity of row i of H_Z on the final readout, i.e. to the augmented observable H_i. Hence an undetected fault set has images whose H parts sum to zero, and whose L parts sum to its logical effect. Its nonzero images therefore sum to an extended-code logical; cancelling equal pairs leaves a set of distinct columns of weight at most the number of faults. If the set has eleven faults and $d_{\text{ext}} = 11$, the eleven images are nonzero, distinct and form one of the listed weight-11 logicals. The template has no X-check detectors in a Z-basis memory; adding them would only remove undetected fault sets, so the bound also holds with all detectors.

Computed check. For every circuit and every R used in S6 we confirmed both facts on the DEM in both bases: each mechanism's per-check detector parity equals the H part of its image, its logical effect equals the L part, and the set of images is exactly the 360 extended columns.

S6. COMPUTATION FOR LEMMA 3

By S5, a weight-11 undetected logical selects, for each of the eleven columns of one listed extended logical, one DEM mechanism with that image, such that the detector sums vanish. For each of the 4224 logicals per basis, a CP-SAT model has a Boolean variable per mechanism of the eleven image groups, an exactly-one constraint per group, and an XOR constraint per detector. Infeasibility for all 4224 logicals at a given R means that no weight-11 undetected logical exists at that R.

Results: circuit A, $R = 1, \ldots, 32$, both bases: all infeasible. A second, independently written lift program agrees and extends the range to $R = 36$. Circuit B: $R = 1, \ldots, 24$; circuit C: $R = 1, \ldots, 14$ and 16, plus a solver-free meet-in-the-middle check (split the eleven groups in two halves, enumerate all detector sums of each half, intersect) for $R \leq 3$; all infeasible. Positive control: on Fig. 12 the same programs find a realisation for all 4224 logicals.

Certificates. For circuits A ($R \leq 32$), B ($R \leq 24$) and C ($R \leq 14$ and 16), both bases and all 4224 logicals, the lift instances were encoded as CNF. Instances were combined into batches by a selector disjunction: the batch formula is the disjunction of its members, each guarded by its own selector variable, with no variables shared between members, so that unsatisfiability of a batch implies that of each member. Each batch was refuted by CaDiCaL; each refutation is a DRAT proof verified with drat-trim, whose LRAT core was verified with the formally verified checker cake_lpr. As a positive control, every one of the 4224 Fig. 12 instances at $R = 1$ is satisfiable and decodes to a valid lift.

S7. PROOF OF LEMMA 4

Notation: period $D = 18$; check q has start s_q, instance j is reset at $s_q + jD$ and measured at $s_q + jD + \Delta_q$ with $\Delta_q \leq D - 1$; $\Delta = \max_q \Delta_q = 17$; $S = \max_q s_q - \min_q s_q = 5$. Faults are attached to layers.

(1) Settling. If all faults of a set $F_<$ lie at layers $\leq a$, the data frame they produce is constant from layer $a + D - 1$ on: only an ancilla fault can later change the data, through the remaining CNOTs of its own instance, all at layers $\leq a + D - 2$; a freshly reset ancilla never pushes errors onto the data (an X-check ancilla only picks up Z errors from the data, a Z-check ancilla only X errors). Every instance reset after $a + D - 1$ is fault-free and measures $H_c e$. If all faults of $F_>$ lie at layers $\geq b$, measurements before b are unaffected.

(2) Splitting. Let $F = F_< \cup F_>$ be a weight-11 undetected logical with no fault strictly between a and b. A comparison detector (c, j) responds to $F_<$ only if $s(c, j-1) < a + D - 1$ and to $F_>$ only if the measurement of instance j, at $s(c, j-1) + D + \Delta_c \leq a + 2D + \Delta - 2$, is at a layer $\geq b$. A first-round detector $(c, 0)$ responds to $F_>$ only if $b \leq s_c + \Delta_c \leq 21$, so with $a \geq 0$ it responds to both parts only if $b - a \leq 21 < G$. A final detector responds to both only if $b - a < D + S + \Delta$. Hence for $b - a \geq G = \max(2D + \Delta - 1,\ D + S + \Delta) = 52$ no detector responds to both parts; each part is undetected, their logical effects add to a nonzero value, and one part has weight ≤ 10, contradicting Lemma 1. Consecutive faults of F are therefore less than G layers apart, and F spans at most $10(G-1) = 510$ layers.

(3) Deletion. The noisy SD6 circuit with R rounds equals the circuit with $R - 1$ rounds with an 18-layer block inserted at layer x, for every $x \in [4, 18(R-1)]$ (checked layer by layer in both bases; the template is exactly periodic). Removing a block $[x, x + D)$ that contains no fault of F removes one measurement of every ancilla; a detector of the shorter circuit that does not straddle the cut equals a detector of the longer one, and one that straddles it equals the sum of the two detectors around the removed measurement. F therefore stays undetected, with the same logical effect, provided the frame entering layer $x + D$ of the longer circuit equals the frame entering layer x of the shorter one; this holds if $x + D \leq f_1$ (all zero; f_1 the first fault layer) or $x \geq f_{11} + 2D - 2$ (data frame settled by (1), all ancillas in flight fault-free). *(4) Counting.* Write $f_1 \leq f_{11}$ for the first and last fault layers of F; by (2),

$f_{11} - f_1 \le 10(G-1) = 510$. *Front:* while $f_1 \ge 4 + D = 22$, the block $[4, 22)$ is fault-free and can be deleted at $x = 4$, shifting F by -18 layers; afterwards $f_1 \le 21$ and hence $f_{11} \le 21 + 510 = 531$. *Back:* a block can be deleted at $x = f_{11} + 2D - 2$ as long as this x is an admissible insertion point, $x \le 18(R-1)$. When no further deletion is possible, $18(R-1) < f_{11} + 2D - 2 \le 531 + 34 = 565$, so $R - 1 \le 31$ and $R \le 32$. (Using $2D - 1$ in place of $2D - 2$, as a more conservative variant does, gives the same bound; the threshold $2D - 3$ would already suffice.) Thus every weight-11 undetected logical of a circuit with R rounds yields one of a circuit with at most 32 rounds, and with Lemma 3 none exists for any R.

For circuits that alternate with their time reverse, Shaw and Terhal reduce any number of rounds to a single round [17]; the gap argument above needs no such symmetry.

Circuits B and C. For circuit C, rounds are separated in time; a fault-free round inside the span of F splits it as in (2), so F occupies at most 11 consecutive rounds, and periodicity reduces any R to $R \le 13$. For circuit B we use a slab version of (2): a fault-free slab from the earliest reset to the latest measurement of one round splits F, provided that on every data qubit every propagating CNOT of round k inside the overlap with round $k + 1$ precedes every reading CNOT of round $k + 1$. We verified this condition for all rounds of the circuits with $R = 3, 8, 24$ in both bases; it reduces any R to $R \le 23$.

S8. UPPER BOUND

For each basis we found a weight-12 logical operator of the gross code of the corresponding type with CP-SAT and applied it as single-qubit flips immediately after data initialisation; each flip is one SD6 mechanism. For $R = 1, 3, 6, 10$ and both bases (and, for circuit A, also at $R = 2$) no detector fires and the logical observable flips; omitting any one of the twelve flips makes the error detectable. Hence $d_{\mathrm{circ}} \le 12$ for circuits A, B and C.

S9. THE $[[90, 8, 10]]$ CODE: BOUND AND THEOREM 2

Code and circuit. $\ell = 15$, $m = 3$, $x = S_{15} \otimes I_3$, $y = I_{15} \otimes S_3$, $A = x^9 + y + y^2$, $B = 1 + x^2 + x^7$, $H_X = [A \,|\, B]$, $H_Z = [B^T \,|\, A^T]$ [1]; $n = 90$, $k = 8$, and an exhaustive search finds no logical of weight ≤ 9 and 558 X-type logicals of weight 10, so $d = 10$. Labels: $A_1 = x^9$, $A_2 = y$, $A_3 = y^2$, $B_1 = 1$, $B_2 = x^2$, $B_3 = x^7$; qubits and checks are indexed by $i = 3a + b$, and the check at (x, y) acts as in S1. The 45 translations are code automorphisms that preserve the labels, and the circuit of Table I is invariant under all of them. It is legal (each data qubit takes part in at most one CNOT per step; every check's six CNOTs cover its support at distinct steps inside its window) and deterministic (each of the 405 overlapping pairs of X and Z checks shares an even number of qubits on which the X CNOT precedes the Z CNOT). X-type CNOTs occupy steps $\{1, 3, 5, 6, 8, 10\}$ and Z-type CNOTs steps $\{2, 4, 6, 7, 9, 12\}$, so each data qubit's cyclic sequence is $XZXZXZ$ and no partition of a round into an X part and a Z part exists.

Every choice of orders has $d_{\mathrm{ext}} \le 8$. Up to s_{12}, s_{56} and reversal, a check of this code has 90 order types; for 36 of them the extended code of that check alone already has a logical of weight ≤ 8. For the remaining 54, consider each of the 558 weight-10 X-type logicals l: any realised residual columns inside l whose total size reduction is at least 2 (one triple or quadruple, or two disjoint pairs) give, with the rest of l, an extended logical of weight ≤ 8. We encode the choice of one admissible type per check as a SAT instance with one clause for each such combination (17910 combinations, each verified explicitly as an extended logical of weight ≤ 8; 3555 variables and 21690 clauses), in a form that only strengthens the statement (types are required "at least one" per check, and realisation is a one-way implication). CaDiCaL reports unsatisfiability and the DRAT proof is verified by drat-trim. Hence for every assignment of orders the X-side extended code has a logical of weight ≤ 8, and every non-interleaved circuit has $d_{\mathrm{circ}} \le 8$ by SBB's Cor. 1. A non-interleaved circuit whose extended code equals that of Table I lifts all 585 weight-8 extended logicals at $R = 1$, so 8 is attained.

Lists. The extended code of the orders of Table I has 225 merged columns in each basis (5 orbits under the 45 translations). There are no extended logicals of weight ≤ 7, exactly 585 of weight 8 and 16830 of weight 9 in each basis; a depth-first search and an independent meet-in-the-middle enumeration give identical sets.

Lift. Following Lemma 2, we decide for every weight-8 extended logical whether it lifts, with or without one additional zero-image DEM mechanism, and for every weight-9 extended logical whether it lifts. Zero-image mechanisms with detectors are, in every round, exactly the 45 ancilla readout flips (checked for $R = 1, \ldots, 12$); all zero-image mechanisms of the DEM are admitted as candidates, so this description is not used. For $R = 1, \ldots, 12$ and both bases no instance lifts (full counts, no early stopping), with three implementations: a frontier dynamic program, a CP-SAT model, and a per-logical SAT model; at $R = 1$ an exhaustive enumeration of the products agrees. Controls: a nearby

interleaved circuit with $d_{\mathrm{circ}} = 9$ gives 315 weight-9 lifts at $R = 1$ and $R = 2$; the non-interleaved circuit above gives 585 weight-8 lifts, 540 weight-8 lifts with a zero-image fault, and 16830 weight-9 lifts at $R = 1$. The image property (S5) and the telescoping identity were asserted on the DEM for every R used.

All R. In round j all CNOTs lie in layers $16j+1, \ldots, 16j+12$ and all measurements in layer $16j+15$, and the noisy circuit with $R+1$ rounds equals the R-round circuit with one 16-layer block inserted (checked for $R = 3, \ldots, 14$, both bases). If a minimal undetected logical has a fault-free round strictly inside its span, the faults before that round form an undetected set by themselves (from that round on every measurement reads the same syndrome, consistent with the final readout), so both parts are undetected and one of them is a lighter logical. Hence a weight-9 logical spans at most 9 rounds, and deleting fault-free rounds after the second and before the last maps it into a circuit with at most $2 + 9 + 1 = 12$ rounds.

Upper bound. A weight-10 data logical, applied as flips after data initialisation, fires no detector and flips the observable for $R = 1, 2, 3, 6, 10, 13$ in both bases.

S10. DIRECT SEARCH ON THE DETECTOR ERROR MODEL

The search looks for a minimum-weight set of DEM mechanisms with zero detector sum and nonzero logical effect. It branches on the mechanisms covering an odd detector, excluding earlier siblings so that the branches partition the search space; it prunes any partial set whose detector sum and logical effect both vanish (such a set cannot be contained in a minimal solution), bounds the remaining weight by a packing of detectors with disjoint covering mechanisms, and closes the last level by table look-up. Root cases are split by the orbits of the translations that preserve the detector set and the logical classes (checked by rank conditions). This is the same family of methods as the extended-code search of S4, implemented independently with different pruning; its $R = 1$ results were reproduced by a second, independently written implementation, which also reproduced $d_{\mathrm{ext}} = 11$ and the list of 4224 logicals of S4, and both were tested on 1672 random instances with known answers. The searches produce no proof certificates.

Results. Gross circuit A: at $R = 1$ there is no undetected logical of weight ≤ 11 in either basis and one of weight 12 is found; at $R = 2$ (66 root cases) and $R = 3$ (96 root cases) there is none of weight ≤ 11, and the injection of S8 gives 12. $[[90, 8, 10]]$ circuit: for $R = 1, \ldots, 12$ there is none of weight ≤ 9 in either basis, and weight 10 is found for $R = 1, 2, 3$; the interleaved control circuit gives 9. For the $[[90, 8, 10]]$ circuit, take a minimal undetected logical of weight $w \leq 9$ at any R: by the splitting argument of S9 it spans at most w rounds and maps into a circuit with at most $w + 3 \leq 12$ rounds, contradicting the search. This proves Theorem 2 for every R without extended-code input; it uses the separation and periodicity checks of S9. At $R = 1$ a SAT formulation of the same question on the DEM is also unsatisfiable in both bases.

S11. FIG. 12 OF SBB: $d_{\mathrm{circ}} = 11$ AND RETIMINGS

$d_{\mathrm{circ}} = 11$. Fig. 12 is non-interleaved, so SBB's Prop. 3 and Cor. 1 give $d_{\mathrm{circ}} = d_{\mathrm{ext}}$ for every R; $d_{\mathrm{ext}} = 11$ by S4. The weight-11 extended logicals are realised by eleven circuit faults (checked by injection for $R = 1, 2, 3$ in both bases).

Lift structure. All 4224 weight-11 extended logicals per basis lift to undetected circuit errors of Fig. 12 ($R = 1, 2, 3$).

Retimings with period ≤ 8. Consider periodic schedules with period $P \leq 8$ shared by all checks, one reset, six CNOTs at distinct offsets and one measurement per check and period, and no data qubit in two CNOTs at one step. Weight 6 forces $P = 8$ and all offsets; keeping the Fig. 12 orders leaves one start per check. A backtracking search without symmetry assumptions finds exactly 32 conflict-free assignments, 4 up to global translation (two SAT encodings agree); all four are colour-uniform, deterministic and non-interleaved, and one is Fig. 12 itself. For each of them and every $R = 1, \ldots, 6$ in both bases we injected a weight-11 error that fires no detector and flips the observable.

Longer periods (bulk). For longer periods we use a Pauli-frame model of the bulk. A data X error at time θ on qubit w is first seen by Z check c (readouts at $T_c + nP$) in instance $H_c = \min\{n : T_c + nP > \theta\}$, i.e. for $\theta \in (T_c + (H_c - 1)P,\, T_c + H_c P)$. A weight-11 logical is *model-liftable* if instance indices H_c can be chosen such that the errors seen by each check can be paired, both errors of a pair being seen in the same instance (so all comparison detectors cancel), and, for every data error, the intervals of its three checks have a common point. For three open intervals of length P on the line, pairwise intersection, which is a difference constraint $H_c - H_d \leq \lfloor (T_d - T_c)/P \rfloor + 1$, implies a common point (one-dimensional Helly); an SD6 fault on w is available there (the two-qubit channel after the CNOT with the latest left endpoint contains $X_w \otimes I$). A residual of length 2 is deposited by an ancilla fault after the fourth CNOT of its check, which fixes its arrival times. *Direction used.* If the difference constraints of a

logical of ten data errors and one length-2 residual are feasible, the corresponding eleven faults, placed away from the time boundaries, form a genuine undetected logical error; this was derived independently twice and confirmed by 75 direct injections, each firing no detector and flipping the observable. Non-existence statements need exactly this direction: an unsatisfiable search for a schedule in which no logical is model-liftable shows that every schedule has a model-liftable weight-11 logical, and when the search uses only logicals of the proven form, every schedule has a genuine weight-11 undetected error for sufficiently many rounds. The converse (model blocking implies exact blocking) is not used and fails at the initial time boundary, where the exact lift can succeed via first-round detectors. Blocking a model-liftable logical requires a negative cycle in its constraint graph; we encode this for all modelled logicals, for CP-SAT and in a lazy-clause SAT loop. The certificates block every model-liftable logical. Re-running the same encoders with only the 360 logicals per basis of the form ten data columns plus one length-2 residual gives UNSAT for every row of Table S3, each accompanied by a DRAT proof, so each obstruction rests only on the direction proven above. The resulting status for retimings that keep the Fig. 12 orders is given in Table S3. Per-check-free pipelined schedules with $P \geq 9$ and different CNOT orders are not covered; the latter is how the new circuits escape.

TABLE S3. Retimings of Fig. 12 that keep its CNOT orders. "Obstructed": no legal deterministic circuit of the family blocks every model-liftable weight-11 logical. Each row was also re-solved with only the ten-data-plus-one-length-2-residual logicals, which again gives UNSAT; every obstruction is therefore proven.

family	status
24-translation symmetric, common frame, any P	obstructed
24-translation symmetric, pipelined, $P = 8, \ldots, 12$	obstructed
subgroup-invariant pipelined, $P = 9$: three subgroups of order 12, three of order 6, one of order 8, two of order 4 and one of order 3	obstructed
subgroup-invariant pipelined, $P = 10$: one subgroup each of order 12, 8, 6	obstructed
no symmetry, common frame, $P \leq 16$	obstructed (one encoding)
other pipelined families with $P \geq 9$; symmetric pipelined $P \geq 13$	open

S12. SEARCH AND OPTIMALITY

Search. The circuits were found with SAT models over schedules symmetric under the 24 colour-preserving translations. Variables: for each of the six classes, one of the eight orders of the Fig. 12 class, a start and six CNOT times; constraints: legality (six distinct times modulo P per data qubit), determinism (on every overlapping pair of X and Z checks, the number of shared qubits on which the X instance acts before the Z instance is even, for every instance offset), and, added lazily, one clause per frame-model-liftable weight-11 extended logical forbidding the timing that makes it liftable. Models were then certified with S5–S7. With the Fig. 12 orders pinned the model is unsatisfiable, in line with S9. As a control, 233 random legal deterministic schedules of the common-frame family, sampled without the blocking clauses, all have liftable weight-11 logicals.

Minimum period. With the same model and no circuit fixed, pipelined schedules with $P = 8, \ldots, 17$ and common-frame schedules with $P = 20$ are unsatisfiable. Every unsatisfiable clause set in the data set (129 in total: 104 for the obstruction and optimality instances with all logicals, and 25 length-2-only instances for Table S3) is accompanied by a DRAT proof produced by CaDiCaL and verified with drat-trim; the LRAT core was additionally verified with cake_lpr (SM S14). For the pipelined case a second, independent encoding (absolute times, one-sided difference literals) agrees for $P = 8, \ldots, 17$. The final clause sets were re-solved from scratch by independent SAT solvers (three for $P = 20$ common frame and $P = 12$, two for $P = 17$). Completeness of the family encoding: at $P = 8$ it enumerates exactly the 944 legal deterministic circuits found by an independent backtracking search, and 75 independently sampled legal deterministic circuits at $P = 12, 17$ (pipelined) and 20 (common frame) are all models. Soundness of the clauses: in 991 sampled cases, including logicals with several residual columns, every lift asserted by the frame model was confirmed on the exact DEM, and in 99 sampled cases every violated clause corresponded to an exact lift. Unsatisfiability means that every legal deterministic circuit of the family has a model-liftable weight-11 logical. For logicals of ten data errors and one length-2 residual this implies a genuine weight-11 undetected error for sufficiently many rounds (S9); for logicals with several residual columns the same implication is confirmed in the sampled cases above but not proven in general. The minimum periods are therefore established within the model and refer to sufficiently many rounds. Restricting the CNOT window of class X_0 or of class Z_2 to at most 14 steps is unsatisfiable at $P = 24$, while each

of the other four classes can be so restricted; since a window spans at most $P-3$ steps, this suggests why $P \geq 18$ is needed, but the bound itself rests on the direct runs. At $P = 8$ all 944 legal deterministic circuits of the family were also checked directly: each has a weight-11 undetected error at $R = 1$ in the Z basis.

Uniqueness of the order class. Per check, the extended-code columns are the data columns and three residual sets; up to reversal there are 90 types per basis, of which 45 contain a residual of length 3 that, together with nine data columns, forms a weight-10 extended logical. For each of the 24 colourings by cosets of translation subgroups of index 2, 3, 4, 6, 8 and 9 (3, 4, 3, 12, 1 and 1 subgroups), with one type per colour, a SAT search with witness clauses leaves only the translates of Fig. 12 with $d_{\text{ext}} \geq 11$ in both bases; every witness used was re-verified as at most ten distinct columns of that assignment's extended code with vanishing H part and nonzero L part. Colourings of index 12, 18, 24 and 36 and per-check-free assignments are not covered.

Less symmetry. In the same model, pipelined schedules invariant under the order-12 subgroup generated by $(1, 2)$ are obstructed for every $P \leq 11$, and schedules invariant under any subgroup of order ≥ 2 are obstructed at $P = 8$. Schedules without symmetry, and the remaining subgroups at $P = 12, \ldots, 17$, are open.

S13. SENSITIVITY TO IDLE NOISE

Noise model. Starting from SD6 we multiply only the idle depolarizing channel by λ, $p_{\text{idle}} = \lambda p$; $\lambda = 1$ reproduces SD6 exactly and $\lambda = 0$ coincides with circuit-level depolarizing noise without idle errors (both checked on the detector error models). The CX-native variant uses SI1000 coefficients with CX, RX and MX treated as native operations; it is not the standard SI1000 model. Compiling CX to $H\,\mathrm{CZ}\,H$ and using Z-basis resets and measurements would raise the per-round fault mass of the period-18 circuit by about 12% and that of the period-8 circuits by 6–9%, so the variant slightly favours the period-18 circuit.

Direct sampling. $R = 12$, identical BP-OSD and Relay-BP decoders with exact priors, both bases. Table S4 lists per-round ratios of the logical error rate of the period-18 circuit to Fig. 12 and to the IBM circuit, with 95% intervals. At $\lambda = 0$ and $\lambda = 0.1$, $p = 4 \times 10^{-3}$, an independent re-sampling with a different noise implementation and BP-OSD code gave 0.37 [0.28, 0.49] and 2.73 [2.14, 3.47] for the ratio to Fig. 12. Several entries rest on few failures; at $p = 2 \times 10^{-3}$ and $\lambda = 0$ no circuit can be distinguished.

TABLE S4. Per-round logical error rate of the period-18 circuit divided by that of Fig. 12 and of the IBM circuit (95% intervals; "–": too few failures).

noise	p	to Fig. 12, BP-OSD	to Fig. 12, Relay-BP	to IBM, BP-OSD	to IBM, Relay-BP
$\lambda = 0$	4×10^{-3}	0.34 [0.19, 0.59]	0.25 [0.11, 0.60]	0.89 [0.45, 1.7]	1.0 [0.33, 2.9]
$\lambda = 0$	3×10^{-3}	0.40 [0.16, 1.02]	0.40 [0.09, 1.9]	1.0 [0.34, 3.0]	0.50 [0.11, 2.4]
$\lambda = 0.1$	4×10^{-3}	3.0 [1.9, 4.7]	2.4 [1.3, 4.5]	3.7 [2.2, 6.0]	7.8 [2.9, 20]
$\lambda = 0.1$	3×10^{-3}	2.5 [0.81, 7.5]	4 [0.56, 22]	3.3 [0.96, 11]	–
$\lambda = 0.1$	2×10^{-3}	2.3 [1.09, 4.95]	–	1.1 [0.19, 4.9]	–
$\lambda = 0.3$	4×10^{-3}	33 [22, 50]	73 [42, 124]	44 [28, 68]	164 [79, 339]
$\lambda = 0.3$	3×10^{-3}	15 [6.9, 33]	20 [7.6, 51]	13 [6.3, 27]	–
CX-native	2×10^{-3}	23 [13, 39]	40 [22, 74]	36 [19, 70]	44 [23, 83]

Per-layer rates. Dividing per-round rates by the period treats every layer as having unit duration, consistent with SD6 but not with physical time. Each round of the period-18 circuit has 12 CX-only layers and 6 layers with measurements or resets; Fig. 12 has 2 and 6. If measurement and reset layers are much longer than CX layers, the ratio of round durations falls from 2.25 towards 1.1–1.3, and the per-layer advantage of the longer period largely disappears.

Leading order. For an ideal decoder at low p, the per-round rate is dominated by weight-12 logical errors with total weight $A_6 p^6$. Summing over the enumerated families (weight-11 logicals and one type of weight-12 logicals), the ratio $A_6(\text{period } 18)/A_6(\text{Fig. } 12)$ is 0.0104 at $\lambda = 0$, 0.084 at $\lambda = 0.1$, 0.84 at $\lambda = 0.3$ and 26 at $\lambda = 1$, crossing 1 near $\lambda \approx 0.32$. Caveats: A_6 is a union-type estimate, not a bound; the other type of weight-12 logical is not enumerated for either circuit, and an unknown factor F in their relative weight moves the crossing (to 0.50 for $F = 0.3$, 0.24 for $F = 2$, 0.20 for $F = 3$); the estimate describes $p \lesssim 10^{-4}$, not the sampled range (at $\lambda = 0.3$ it predicts near equality while direct sampling at $3\text{–}4 \times 10^{-3}$ shows a factor 15–73); and it says nothing about practical decoders.

S14. VERIFICATION AND DATA

All circuits were built and simulated with Stim [16]; exact feasibility questions were decided with CP-SAT and with CDCL SAT solvers, and the extended-code searches use a dedicated depth-first search. The following were obtained with two independent implementations that agree: the construction of the extended code and $d_{\text{ext}} = 11$; the list of 4224 weight-11 logicals; the lift computation of Lemma 3; the enumeration of period-8 schedules; and the minimum-period runs. The arguments of Lemmas 1 and 4 were checked independently, and their structural premises (template identity, image set, periodicity, detector structure) were verified by computer for every circuit. The data set (Data availability) contains the circuit specifications (circuits A, B, C and Fig. 12), the lists of 4224 weight-11 extended logicals for each basis, the witnesses (the weight-12 upper-bound witnesses and the weight-11 witnesses for Fig. 12 and its retimings), the certificates (clause sets of the unsatisfiable instances), and the reproduction scripts for the lower-bound computations; it is available at [Zenodo DOI]. Every unsatisfiable clause set in the data set (129 in total: 104 for the obstruction and optimality instances with all logicals, and 25 length-2-only instances for Table S3), and every batch of the lift computation, is accompanied by a DRAT proof produced by CaDiCaL and verified with drat-trim; the LRAT core was additionally verified with cake_lpr. Here "accompanied" means: proofs of up to 100 MB are included in the data set; larger ones are identified by their SHA-256 hash and can be regenerated byte-identically with the included script. The certificates cover the solver decisions only; the circuit template and the reduction of Lemma 1 are covered by the pen-and-paper proofs and code-level checks of S5. Limitation: $d_{\text{ext}} = 11$ and the completeness of the list of 4224 weight-11 logicals rest on two independent exhaustive searches; we did not obtain a checkable proof certificate for them (a direct CNF encoding is beyond CDCL solvers already at weight 6).